

Arthur Hafliger

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In depth Descriptions on ArthurHafliger.github.io

Education

University of Central Florida - Bachelor of Science in Mechanical Engineering (3.8 GPA), Math/Robotic Systems Minor - 2028
-Pegasus Gold Scholarship Recipient

Pasco-Hernando State College - Associate in arts (3.7 GPA) – 5/20/25
-Magna Cum Laude Student

Technical Skills

CAD - Solidworks(**CSWP**, CSWA), Onshape

Programming - C, C++(Arduino)

Simulation - MATLAB/Simulink fundamentals

Robotics - Forward and Inverse Kinematics, PID Control, Arduino

Manufacturing - 3D Printing



Relevant Coursework

Dynamics, Principles of Electrical Engineering, Differential Equations, Engineering Concepts and Methods, Intro to C Programming

Project Experience

Custom Servo/Cycloidal Drive - Self Directed

3/1/26 - Ongoing

Solo Project

- Designed and manufactured a 3D printable Cycloidal Drive with a 10 to 1 reduction ratio
- Utilized MATLAB to simulate dynamics of a motor to implement a PID controller to control a PWM Output Signal
- Utilized MATLAB to graph angle accuracy and time to convergence

Miniature Boat design - UCF, Orlando, Florida,

1/1/26 - 5/1/26

Team Project

- Designed and developed a 3D printed Boat Hull capable of supporting a load of approximately 7 lbs
- Full CAD model for every part and full assembly creation using SolidWorks
- All hull parts are held together using M4 screws and brass heat activated inserts
- Utilize Jira for team coordination and project management

2 DOF Control Simulation - Self Directed

2/1/26 - 3/1/26

Solo Project

- Utilized MATLAB to simulate control methods and diagnose shortcomings of "3 DOF Robotic Arm"
- Developed a double pendulum model derived from lagrangian mechanics to simulate gravity and control
- Compared ideal full state feedback control to PD control with gravity compensation

3 DOF Robotic Arm - Self Directed

12/1/25 - 2/1/26

Solo Project

- Developed forward and inverse kinematics using a damped least squares Jacobian solver
- Implemented PID velocity control for manipulation by any 3 dimensional vector
- Designed singularity detection and avoidance system

Radial Aircraft Engine CAD - New Port Richey, Florida,

8/1/23 - 5/1/24

Solo Project

- Demonstrate substantial knowledge in SolidWorks CAD, general part modeling, interpreting drawings, and assembly
- Develop innovative solutions to assembly related problems, ensure functionality of basic/mechanical/advanced mates

Linetracking and Manipulating robot - New Port Richey, Florida,

1/1/24 - 5/1/24

Solo Project

- Develop a complex program in C++ enabling line tracking capabilities, utilization of a manipulating arm, operation of a color sensor
- Utilize SolidWorks for design before physical construction, using Vexx brand electronic components
- Develop a mechanically sound and compact design, including a drivetrain, manipulating arm, and a sorting system

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Leadership Experience

SolidWorks Student Leadership Program - University of Central Florida, Orlando, Florida,
Member

9/1/25 - Ongoing

- Completion of CSWP prep Program

Student Advisory Council - Wendell Krinn High school, New Port Richey, Florida,
Co-Chairman

9/1/23 - 5/1/24

- Participate in meetings with school officials, review budgets and proposed plans for school spending, vote on utilization of spending
- Demonstrate effective leadership for the student body by showing awareness of school needs and asking the necessary questions
- Communicate effectively to my fellow Co-Chairmen

Work Experience

Wawa - Port Richey/Oviedo, Florida,

12/9/23 - Ongoing

Customer Service Associate

- Handle deli operations, manage food preparation while maintaining clean standards, demonstrate effective communication to management and fellow associates
- Collaborate with fellow associates to create effective work schedules, demonstrate effective conflict resolution
- Communicate Directly with Customers to ensure satisfaction

Keke's Breakfast Cafe - Port Richey, Florida,

8/1/23 - 12/1/23

Expeditor

- Communicate and act as a bridge between the front of house and back of house operations
- Maintain order in an inherently chaotic environment
- Speak for fellow staff members to ensure problems are solved

Interests

Aerospace, System Dynamics, Physics, Controls, Embedded Systems